SYSTEM DESIGN

Replication Strategies Visual Guide

Keep copies of data on many nodes: durability, availability, and locality.

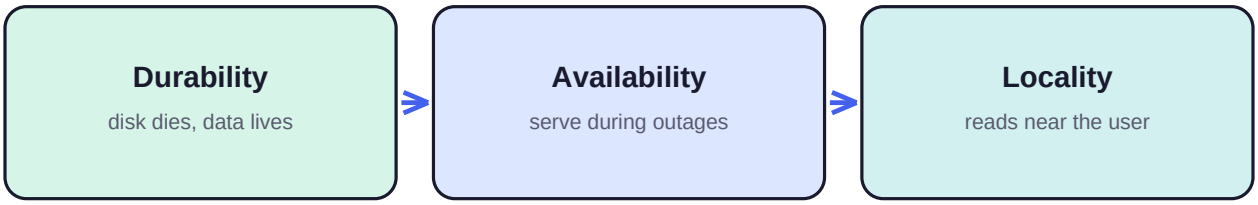
ANIMATED WALKTHROUGH

Step through pages to play the animation

STEP 1/6

Why Replicate

Three reasons, all about surviving reality.



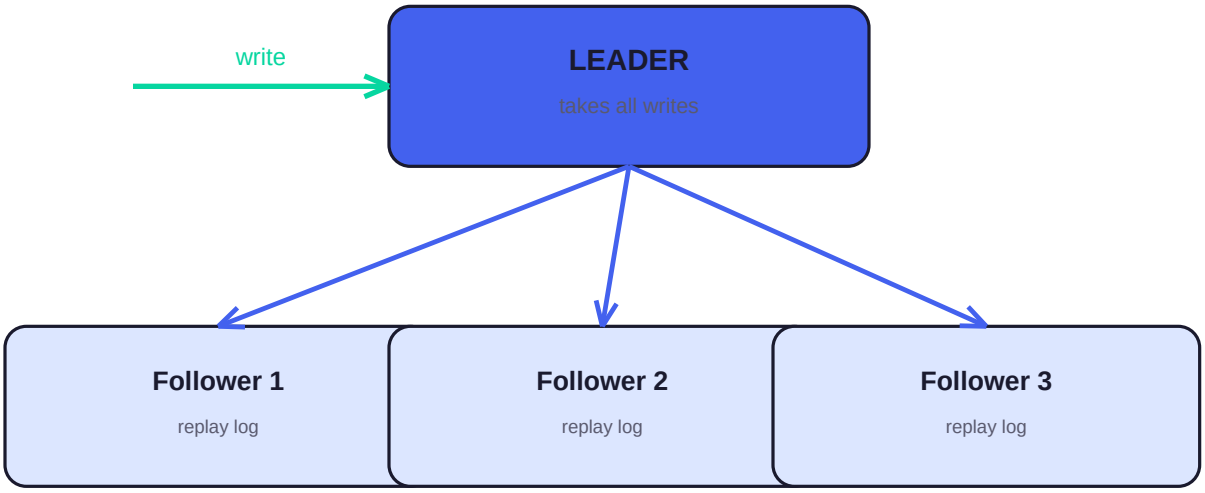
Replication factor (RF): how many copies. RF=3 is the industry default.

- Every strategy below is a different answer to: who takes writes, and how do copies converge?
- More copies = more durability, but every copy is a consistency problem to manage.

STEP 2/6

Leader / Follower (Single Leader)

One writer, many readers. Simple, ordered, the default.

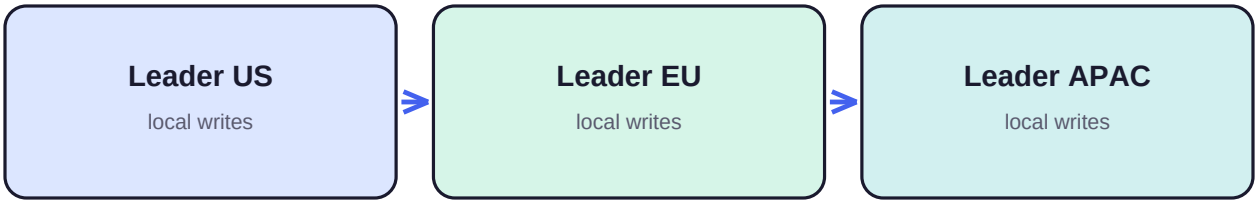


Writes are totally ordered by the leader -- no write conflicts ever. Reads scale by adding followers.

STEP 3/6

Multi-Leader

Write locally anywhere; leaders sync with each other.



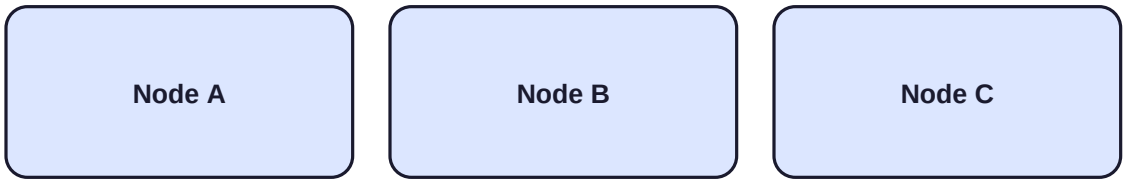
US <-> EU <-> APAC replicate asynchronously. Low write latency on every continent.

- The price: concurrent writes to the same key CONFLICT -- you need resolution (last-write-wins, CRDTs, app merge).
- Topology matters: star, ring, or all-to-all changes failover behavior.
- Use when: offline-first apps, geo-distributed writes, low-latency local writes.

STEP 4/6

Leaderless (Dynamo-Style)

No leader at all: any node accepts the write.



Client writes to any W nodes, reads from any R nodes. No failover -- there is no leader to lose.

- Availability win: the cluster keeps accepting writes even when nodes are down.
- Consistency work moves to the CLIENT: version vectors detect concurrent writes, app merges siblings.
- Cassandra and DynamoDB live here; quorums (next page) decide what 'enough' means.

STEP 5/6

Synchronous vs Asynchronous

Wait for the copy, or trust it will catch up?

Synchronous

leader waits for follower ack■slower, no data loss

Asynchronous

leader acks, replicates after■fast, may lose recent writes

- Semi-sync (the middle path): wait for ONE follower -- bounded loss, modest latency (MySQL semi-sync).
- Chain replication: writes flow leader -> f1 -> f2, reads from the tail -- ordered AND fast reads.
- Durability is a spectrum, not a switch: pick the point your SLA can afford.

STEP 6/6

Read / Write Quorums

The math that makes leaderless replication consistent.

R + W > N => strong consistency

N=3 replicas, W=2, R=2: every read overlaps a write quorum

- W + R > N guarantees read-write overlap: a read quorum always contains the latest write.
- W + W > N prevents write-write conflicts without a leader.
- Tune it: W=1,R=3 (fast writes, slow reads) ... W=3,R=1 (slow writes, fast reads) ... W=2,R=2 (balanced).
- Sloppy quorum: if a replica is down, write to a stand-in and hand it off later.

STEP ✓/✓

Key Takeaways

Replication Strategies — Remember

- Leader/follower: one writer, totally ordered, simple -- the default for most systems.
- Multi-leader: local writes everywhere, but concurrent writes need conflict resolution.
- Leaderless: any node takes writes; quorums (R + W > N) restore consistency without a leader.
- Sync replication buys durability with latency; async buys speed with potential loss.
- Replicate each shard, tune W/R to your read-write ratio, and RF=3 is the sane default.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev